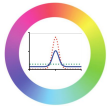




Competition Between Working Memory Precision and Hippocampal Pattern Separation

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INTRODUCTION

✓ Working Memory Precision

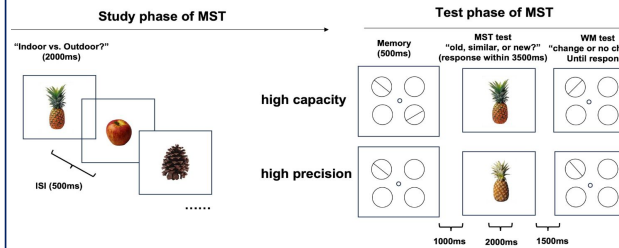
- Quantitative versus qualitative aspects of WM representations; how many items can be remembered versus how detailed the items are
- ✓ **Hippocampal Pattern Separation**
- Process of separating the overlapping inputs into distinct, non-overlapping ones in the hippocampus
- ✓ **Testing Hypothesis**
- WM precision and hippocampal pattern separation rely on the shared cognitive resource

Prediction

- A high load on WM precision will reduce lure discrimination ability, but not on WM capacity
- Increased lure discrimination demands will impair WM performance while there is a high load on WM precision

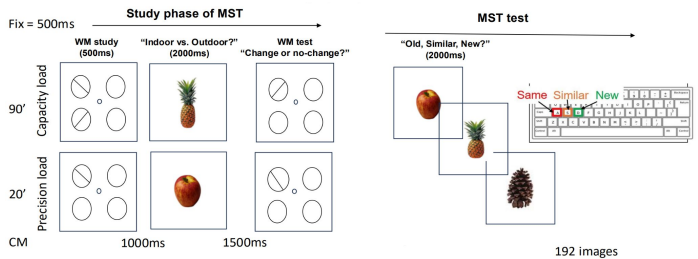
EXPERIMENT

✓ Experiment 1 Task and Procedure (N = 53)



- Study phase of MST is for the **encoding** of a series of objects
- Test phase includes MST with **retrieval** process with a memory test
- It is expected that lure discrimination will decrease during high WM precision load

✓ Experiment 2 Task and Procedure (N = 45)



- Study phase includes MST with **encoding** process
- Test phase includes only **retrieval** process
- It is expected that accuracy in WM precision load will be lower than in WM capacity load

DATA ANALYSIS & RESULTS

✓ Data analyses

- LDI was compared between WM loads: precision and capacity
- WM accuracy was compared across MST trial types (target, lure, and foil in Exp 1, and target vs. lure in Exp 2)

✓ How to calculate lure discrimination (LDI) and WM accuracy?

- Lure Discrimination Index (LDI) is calculated as follows:
$$LDI = p(\text{"similar"}|\text{lure}) - p(\text{"similar"}|\text{foil})$$
- WM accuracy is calculated as the counts of correct responses across all trials

✓ Results

- No significant difference in LDI between WM loads for both experiments
- Experiment 1 Lure condition has a significant effect between loads.

✓ Experiment 1 Results

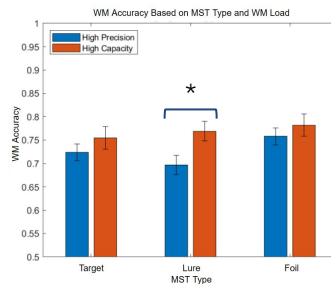


Figure 1. WM accuracy by MST type and precision versus capacity for Experiment 1.

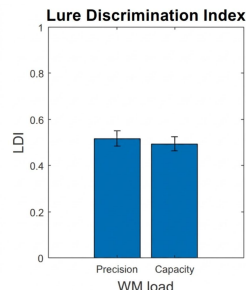


Figure 2. Lure Discrimination Index by WM load for Experiment 1

✓ Experiment 2 Results

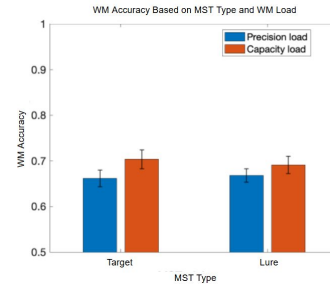


Figure 3. WM accuracy by MST type and precision versus capacity for Experiment 2

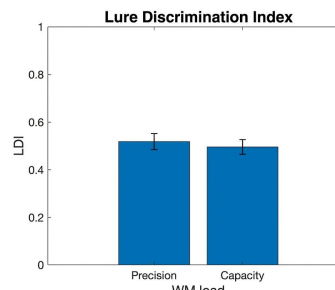


Figure 4. Lure Discrimination Index by WM load for Experiment 2

DISCUSSION

✓ General Discussion

- Competition between WM precision and hippocampal pattern separation may arise during the retrieval of information from long-term memory.
- The results suggest that LTM encoding demands override WM precision as proven by the stability of lure discrimination despite high WM loads.
- We found that lure discrimination ability reduced working memory precision.

✓ What Next?

- Since the competition between WM precision load and hippocampal pattern separation were not tested using neurological methods, more research is needed to if there is a connection on a neurological level.
- Future research could be improved with a larger, more diverse sample size.

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